**Department of Engineering Science**
**School of Science & Technology**




# CES 440 Data Communications
# Ch.4 The MAC Sublayer: Part I

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# OUTLINE

- **Functions of MAC Sublayer**
- **MAC Protocols**
  - ALOHA
  - CSMA
  - CSMA/CD
  - Bitmap, Binary Countdown
  - Limited Contention
  - WDMA
  - Wireless LAN protocols
- **Ethernet**
- **Data Link Layer Switching**
- **Wireless LAN**
- **Broadband Wireless**
- **Bluetooth**

# FUNCTIONS

- **Media Access Control**
  - Determine how to allocate a single broadcast channel (multiaccess channel) among multiple competing users.
  - A sublayer of data link layer (at the bottom of data link layer).
  - Most LANs are using broadcast channels
    - MAC is essential to LANs
- **Classifications**
  - Static channel allocation
    - E.g. FDM: static allocate a portion of the bandwidth to each user.
    - Efficient if each user has heavy traffic load
  - Dynamic channel allocation
    - Dynamically allocate channel to competing users.
    - Efficient for bursty traffics: data coming in irregularly.
      - Most widely used in LANs.

# PROTOCOL: ASSUMPTIONS

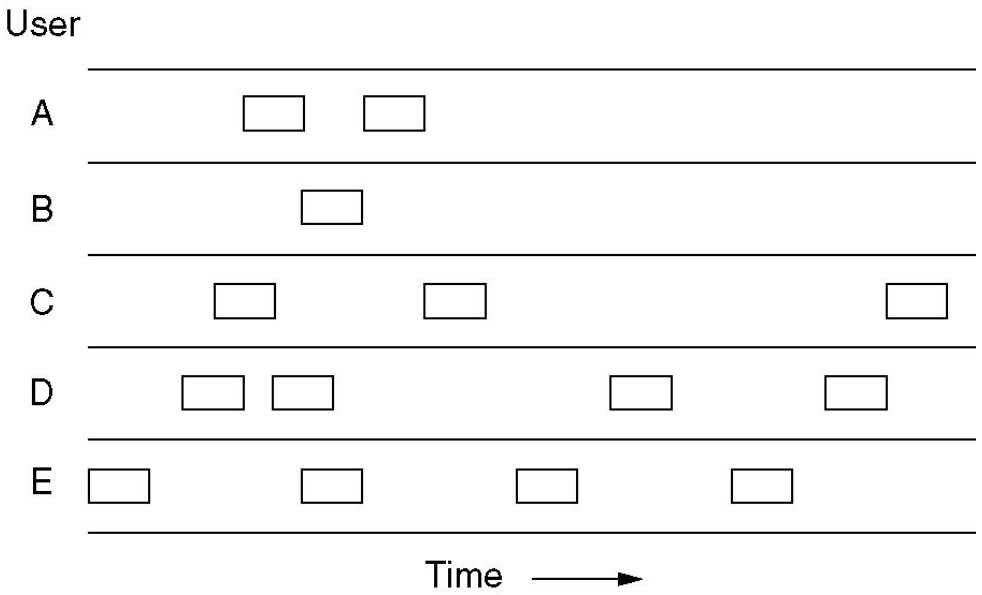
- **Station model**
  - N independent stations.
- **Single channel assumption**
  - One single channel is shared by all users for transmission and reception.
- **Collision**
  - Two frames transmit simultaneously, they overlap in time and collision occurs.
- **Continuous time v.s. Slotted time**
  - Continuous time: frame transmission can begin at any time.
  - Slotted time: frame transmission always start at the beginning of a slot.
- **Carrier sense v.s. No carrier sense**
  - Carrier sense: stations can tell if the channel is in use before transmission.
    - Ethernet
  - No carrier sense: stations cannot sense the channel before trying to use it.
    - Wireless LAN

# PROTOCOL: ALOHA

- **Background**
  - In 1970s, University of Hawaii researchers wanted to enable terminals on remote islands to access main computer in Honolulu.
  - Applicable to system with uncoordinated users competing for a single shared channel.
- **Classification**
  - Pure ALOHA
    - Continuous time
  - Slotted ALOHA
    - Slotted time

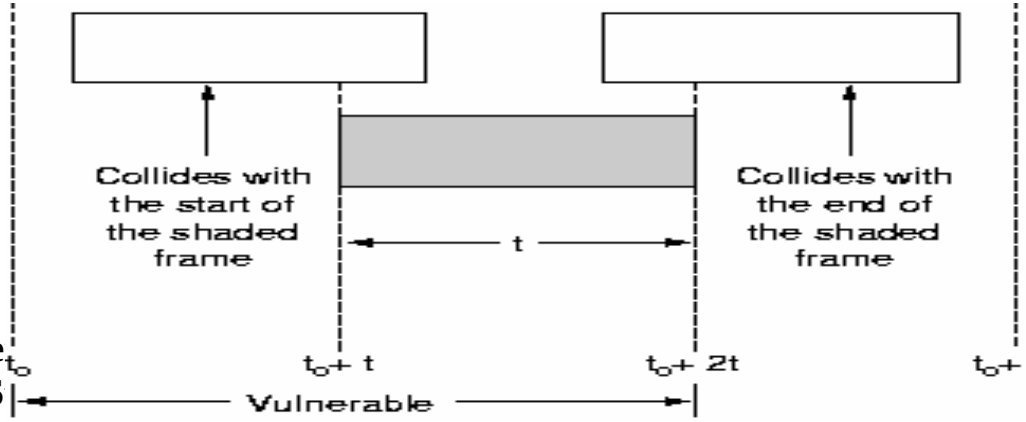
# PROTOCOL: ALOHA

- **Pure ALOHA**
  - Users transmit whenever they have data.
  - Colliding frames are damaged.
  - Sender can find out collision by either listening to the channel or missing of Ack.
  - If collision happens, sender waits for a random time and transmit again.



# PROTOCOL: ALOHA

- **Pure ALOHA Performance**
  - Frame time: the amount of time needed to transmit a frame.
  - Offered load *G*: the average number of frames (new + retransmission) per frame time.
  - Throughput: $S = GP_0$, where $P_0$ is the probability of no collision.
  - Collisions: (assume a frame starts at $t_0 + t$, and the transmission time is *t*)
    - If another frame starts between $t_0$ and $t_0 + t$, it will collide with the shaded frame.
    - If another frame starts between $t_0 + t$ and $t_0 + 2t$, it will collide with the shaded frame.
    - If no other frame is generated between $t_0$ and $t_0 + 2t$ (vulnerable time), no collision!



# PROTOCOL: ALOHA

- **Pure ALOHA Performance (Cont'd)**
  - Assumptions: the # of frames arriving in one frame time follows Poisson distribution with mean *G*. The probability that *k* frames in one frame time is:

$$\Pr[k] = \frac{G^k e^{-G}}{k!}$$

  - The # of frames arriving in two frame times follows Poison distribution with mean *2G*. The probability that k frames in two frame time is:

$$\Pr[k] = \frac{(2G)^k e^{-(2G)}}{k!}$$

  - The probability that no frame in two frame time ( = the probability of no collision)

$$P_0 = \Pr[0] = \frac{(2G)^0 e^{-(2G)}}{0!} = e^{-2G}$$

  - Throughput

$$S = G \times P_0 = G \times e^{-2G}$$

    - E.g. G = 0.5 ➔ S = 0.1839

      G = 1 ➔ S = 0.1353

# PROTOCOL: ALOHA

- **Slotted ALOHA**
  - Times are divided into slots.
  - Computers are only allowed to transmit at the beginning of each slot.
- **Slotted ALOHA Performance**
  - Collisions:
    - The probability of collision happens for a frame is that there are other frames sent out at the same slot ➔ vulnerable time is one frame time *t* ➔ probability of collision is reduced.
  - The probability there is no collision ( = probability no other frame in the vulnerable time)

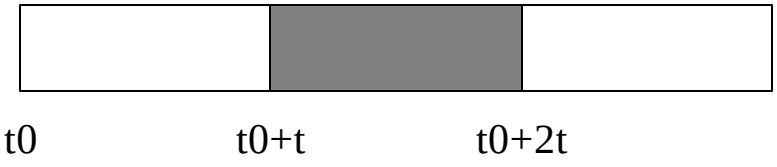
$$P_0 = \Pr[0] = \frac{(G)^0 e^{-(G)}}{0!} = e^{-G}$$

  - Throughput

$$S = G \times P_0 = G \times e^{-G}$$

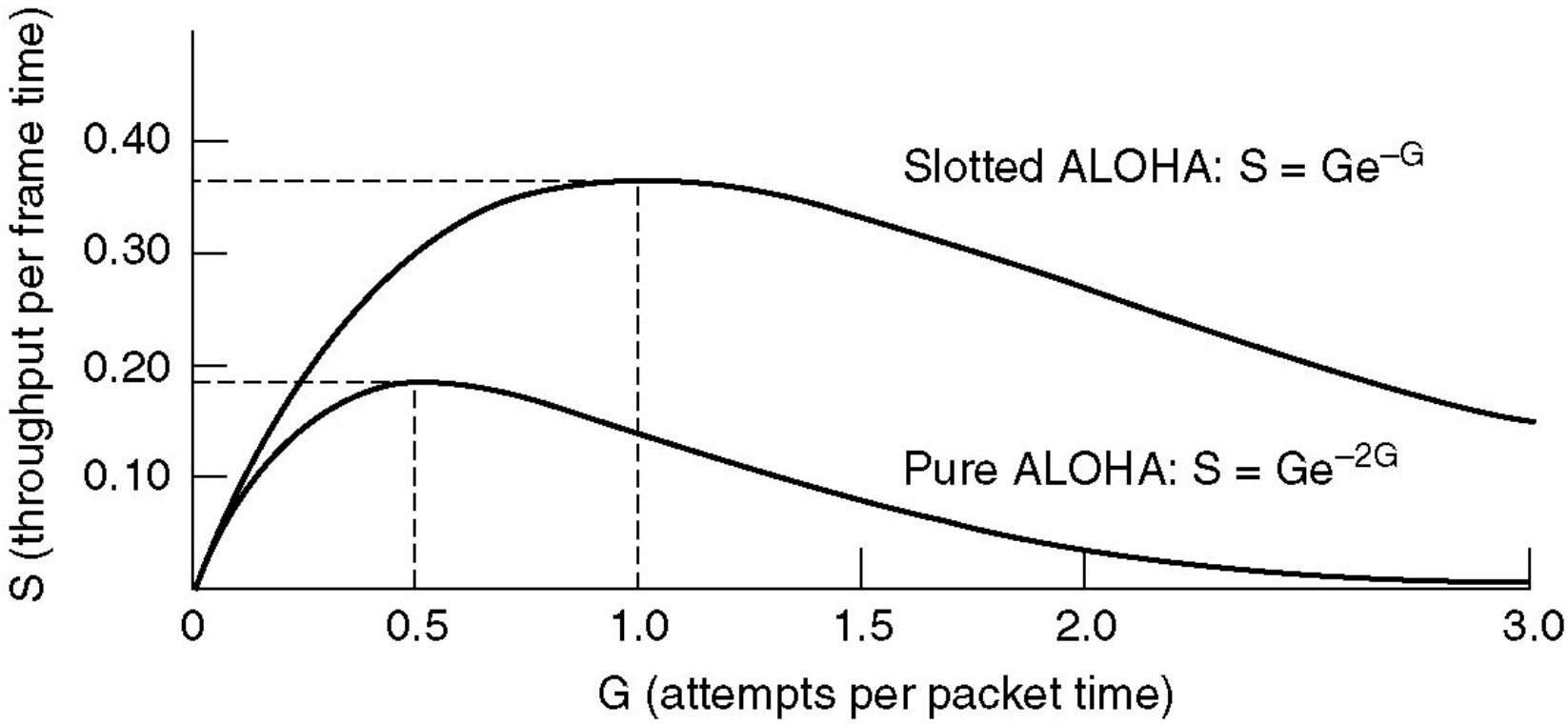
    - E.g. G = 0.5 ➔ 0.3033

      G = 1 ➔ 0.3679



# PROTOCOL: ALOHA

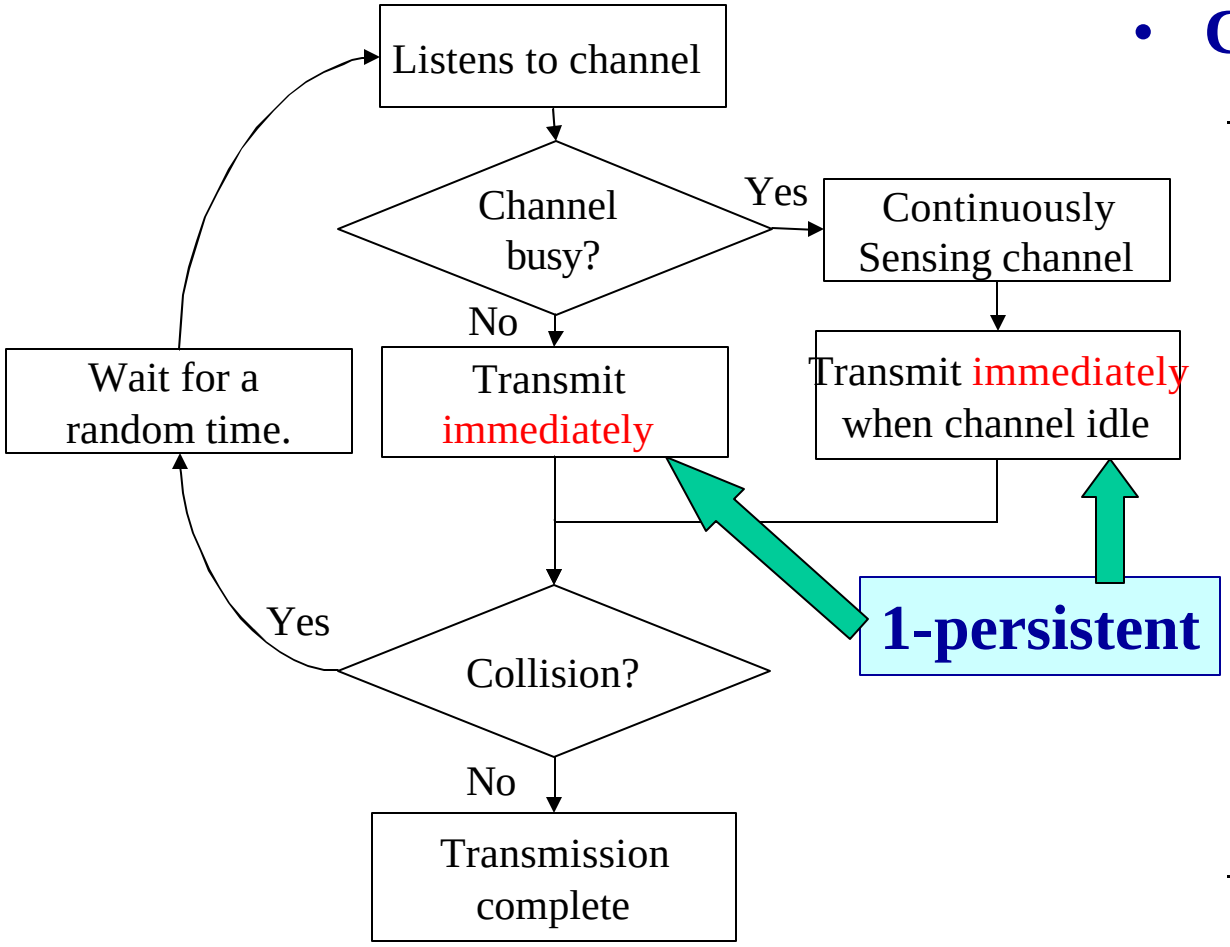
## Throughput v.s. Offered Load



# PROTOCOL: CARRIER SENSE MULTIPLE ACCESS

- **CSMA**
  - Before sending out data, hosts listen to the carrier to see if there is transmission in the channel.
  - If there is transmission, host will back off.
  - If there is no transmission, host will act differently for different CSMA protocols
- **CSMA Protocols**
  - 1-Persistent CSMA
  - *p*-Persistent CSMA
  - Nonpersistent CSMA
  - CSMA with collision detection (CSMA/CD)

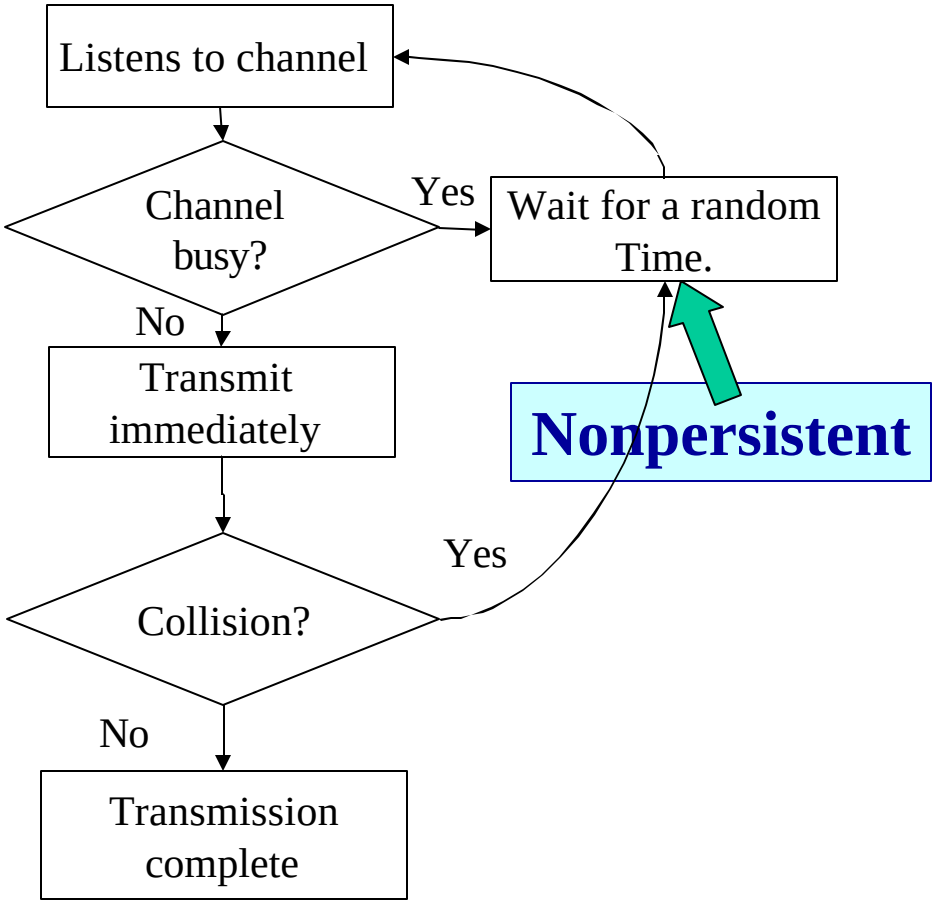
# PROTOCOL: 1-PERSISTENT CSMA




- **Collisions**
  - Case 1: non-zero propagation delay
    - 1) First station starts transmission.
    - 2) The signal has not arrived at the second station.
    - 3) Second station thinks the channel is idle, and starts transmission.
  - Case 2: two stations transmit simultaneously.

- **1-persistent: transmit at probability 1 if the channel is idle.**
- **Due to 1-persistent, probability of collision is high!**

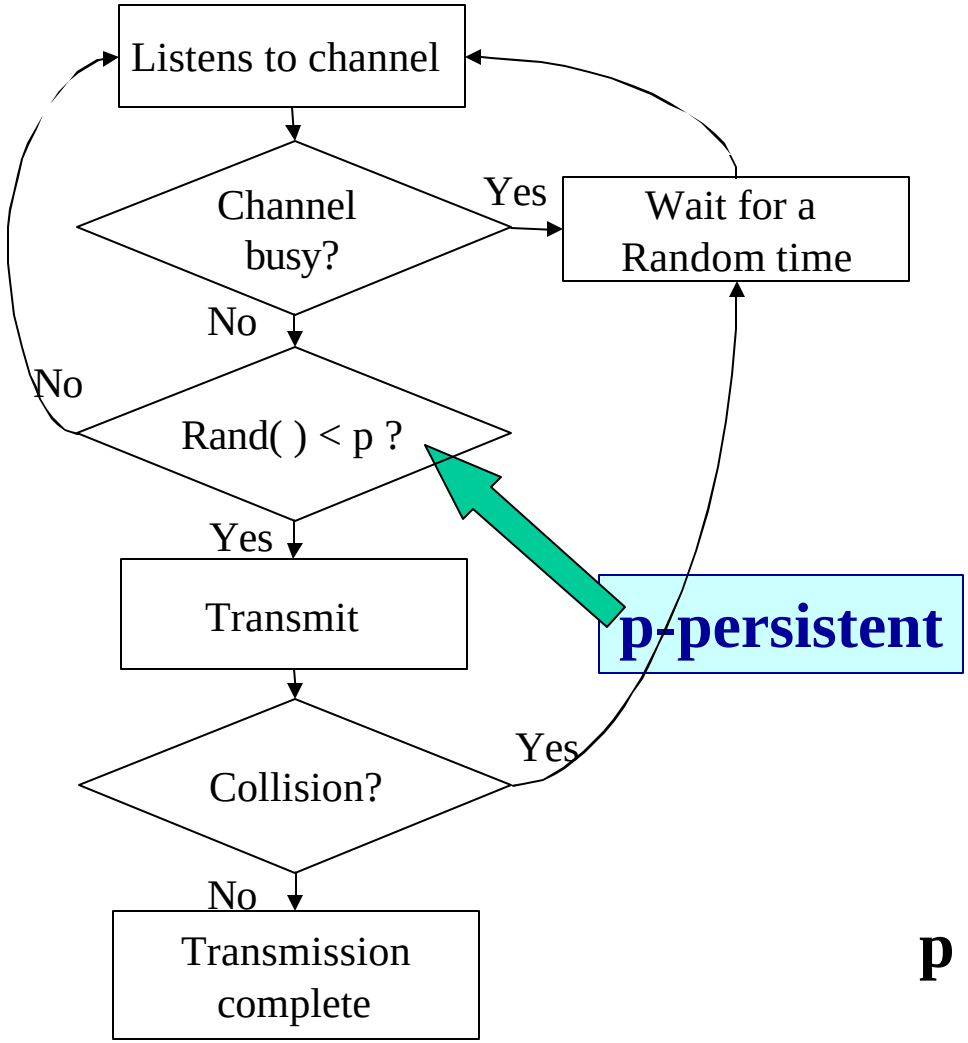
# PROTOCOL: NONPERSISTENT CSMA




- **Compared to 1-persistent**
  - Smaller collision rate.
  - Longer delay.

- **Nonpersistent: sender doesn't persist on transmitting if channel is busy**

# PROTOCOL: P-PERSISTENT CSMA

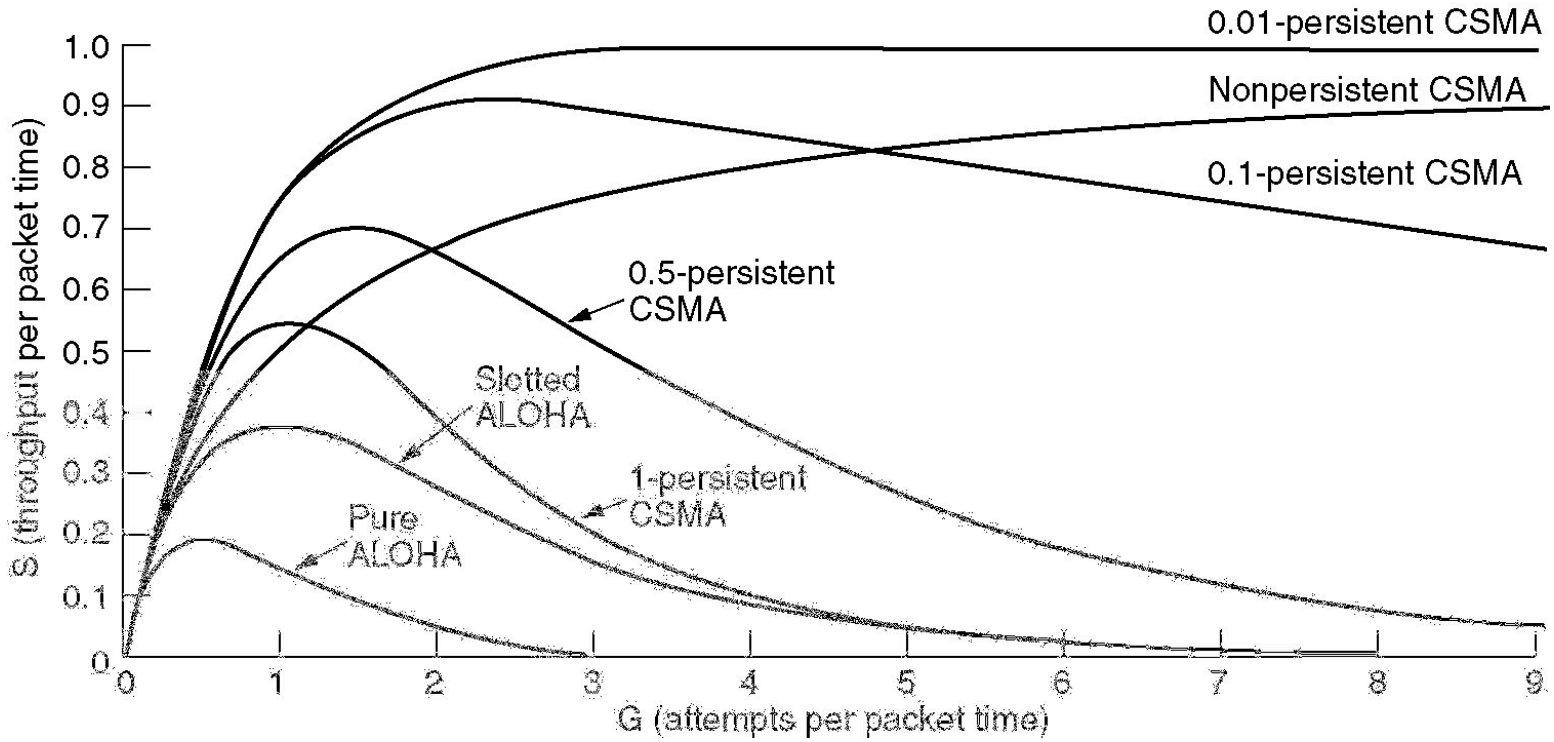



- **Rand( ): generate a random number in [0, 1]**
  - Pr [rand( ) < p] = p
- **p-persistent:**
  - Sender transmits at probability *p* if the channel is idle.
  - Applies to slotted channels.

**p is a number between 0 and 1**

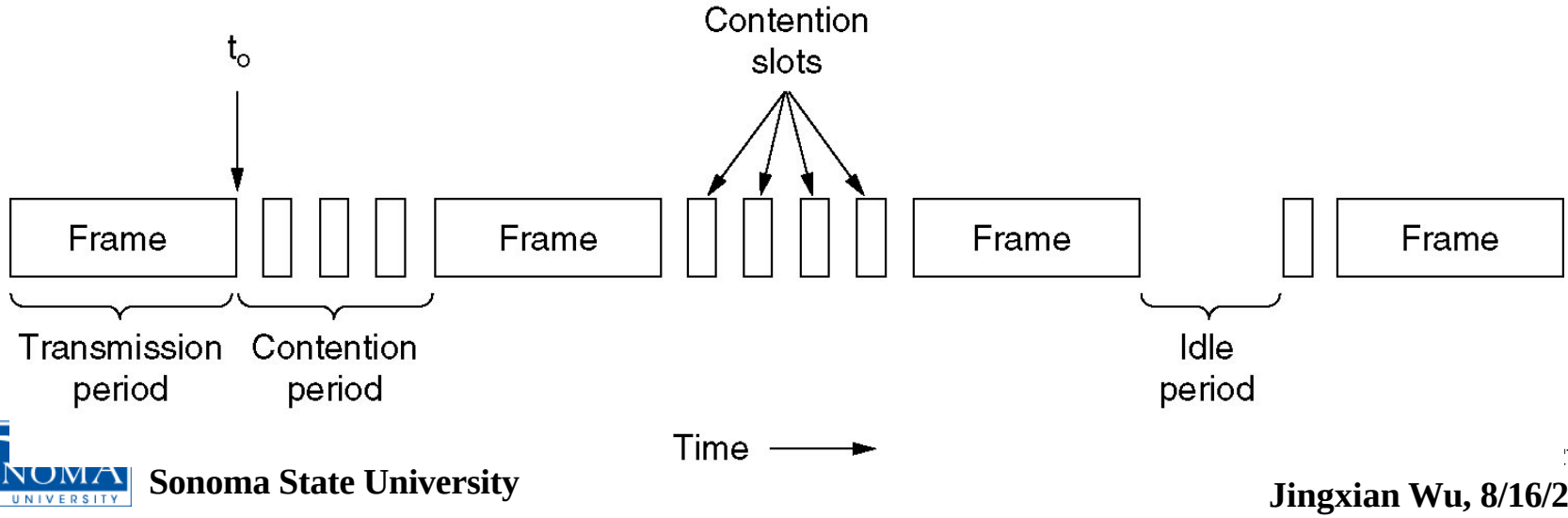
# PROTOCOL: CSMA

## Throughput v.s. Offered Load



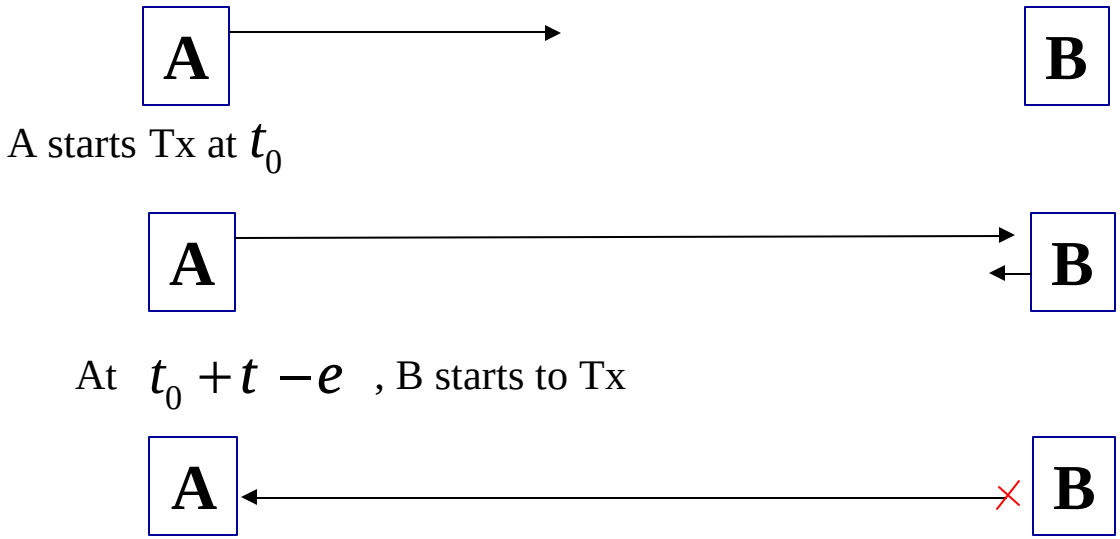
# PROTOCOL: CSMA/CD

- **CSMA/CD**
  - Carrier Sense Multiple Access with Collision Detection
  - Sender sense the channel during transmission
  - The transmission is aborted as soon as a collision is detected ➔ collision detection
    - In regular CSMA system, sender will send the entire frame even though a collision is detected during transmission.
  - Used by most LANs, e.g. Ethernet
  - Three period: contention period, transmission period, idle period.



# PROTOCOL: CSMA/CD

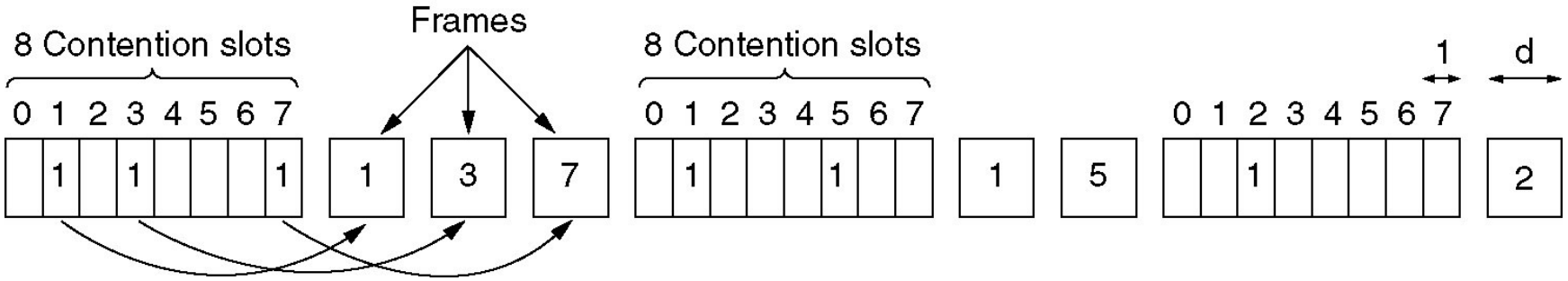
- **One stations transmit at $t_0$, after how long can it decide that there is no collision?**
  - Assume the maximum one way propagation delay is $t$




- **A station cannot be sure that it seized the channel until no collision has been detected after $2t$ since transmission.**

# PROTOCOL: COLLISION-FREE PROTOCOLS

- **Bit-Map Protocols**
  - Assume N stations, each with a unique address.
  - Contention period consists of *N* slots
  - If station *k* has data to send, it puts a '1' in the *k*th contention slot.
  - After the N contention slots passed by, each station has full knowledge of which stations need to Tx.
  - After contention period, stations begin to Tx in numerical order.
  - The desire to transmit is broadcast before actual transmission ➔ reservation protocol.



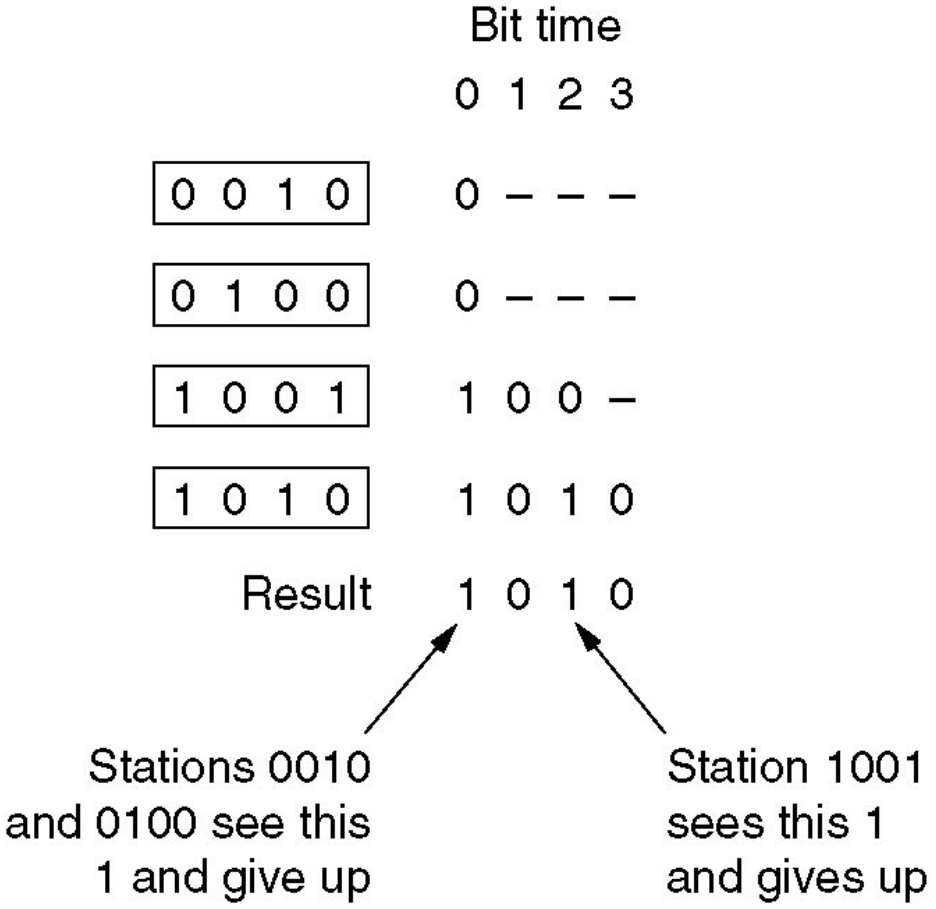
# PROTOCOL: COLLISION-FREE PROTOCOLS

- **Performance of Bit-Map protocols**
  - Assume the average data length is *d*
  - For low load (e.g. only 1 station wants to Tx)
    - *d* bits of data, *N* bits of overhead
    - Efficiency = *d/(d+N)*
  - For high load (e.g. all *N* stations wants to Tx)
    - *d* x *N* bits of data, *N* bits of overhead
    - Efficiency = *dN/(dN+N) = d/(d+1)*

- **Good for systems with high traffic load!**

# PROTOCOL: COLLISION-FREE PROTOCOLS

- **Binary Countdown**
  - Each station is assigned a unique binary address
    - $\log_2 N$ bits are required to distinguish $N$ stations.
  - Assume no propagation delay, and all stations are synchronized.
  - Stations wanting to transmit broadcasting their addresses as a binary string, starting with higher order bits.
  - Perform OR operation in each bit position.
  - Arbitration rule: as soon as a station sees that a higher order bit position that is '0' in its own address has been overridden by '1', it gives up.



# PROTOCOL: COLLISION-FREE PROTOCOLS

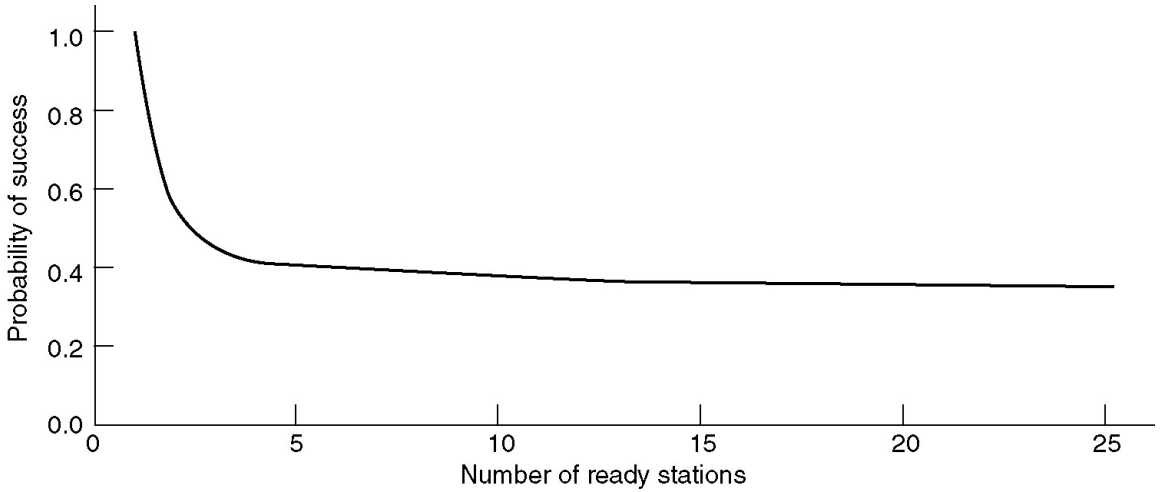
- **Binary Countdown (Cont'd)**
  - Efficiency:
    - Data length: d bits; overhead (address): $\log_2 N$
    - Efficiency: $d/(d+\log_2 N)$
  - Problem: higher numbered stations always have a higher priory than lower numbered stations.
    - Drawback: unfair
    - Advantage: can be used for QoS purposes.
- **Solution for unfairness: Virtual Station Numbers**
  - Stations that just transmit will be given the lowest station number in the next round.
  - E.g. Initially the order is C, D, B, A with C having the highest priority
    - D transmit in the current round
    - In the next round, the order becomes C, B, A, D.

# PROTOCOLS: LIMITED-CONTENTION PROTOCOLS

- **Motivation**
  - Two performance measures: 1. delay, 2. channel efficiency
  - Under light traffic load, contention is preferable
    - Low delay
  - Under high traffic load, collision-free is preferable
    - Contention over high load will lead to lengthy arbitration.
    - Collision-free under high load is more efficient.
- **Can we design a protocol that can adjust the arbitration scheme based on the traffic load?**
  - Thus combine the advantages of Contention and Collision-free protocols
  - Limited-contention protocols.

# PROTOCOLS: LIMITED-CONTENTION PROTOCOLS

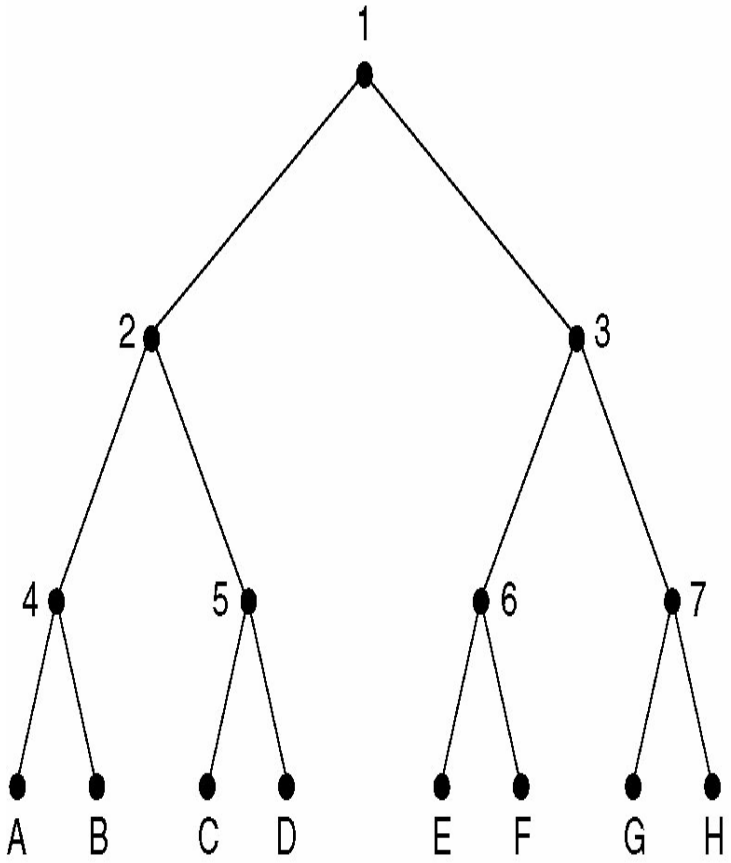
- **Probability of success v.s. Number of ready stations**




- **The probability of success can be increased by decreasing the number of competing stations**
  - Solution: Divide the stations into groups.
  - Only stations within one group are allowed to compete with each other.
  - TDMA among groups ➔ no competition among groups
  - Each group has 1 member: bit-map
  - Only one group: slotted ALOHA

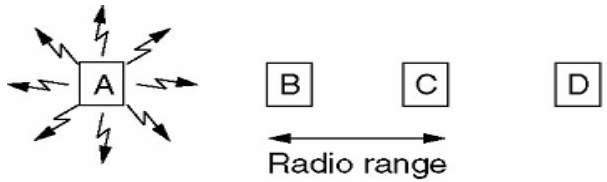
# PROTOCOLS: LIMITED-CONTENTION PROTOCOLS

- **How to dynamically divide groups: The Adaptive Tree Walk Protocol**
  - 1. During contention slot 0, all stations covered by node1 compete.
  - 2. If there is collision, all stations covered by node 2 compete during contention slot 1
    - If success, after transmission, stations covered by node 3 will compete during next contention period.
  - 3. If there is collision at node 2, then all stations covered by node 4 compete at slot 2 ……



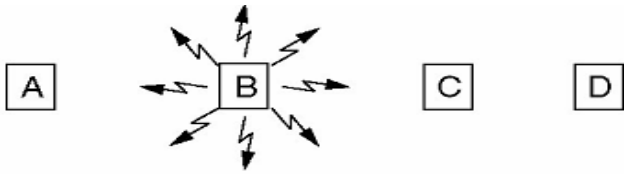
# PROTOCOLS: WIRELESS LAN

- **Can we use CSMA in wireless LAN?**
  - Hidden station problem
    - A is transmitting to B, C couldn't detect signal from A




(a)

  - Exposed station problem
    - C wants to transmit to D, it falsely concludes that carrier is busy by listening to B

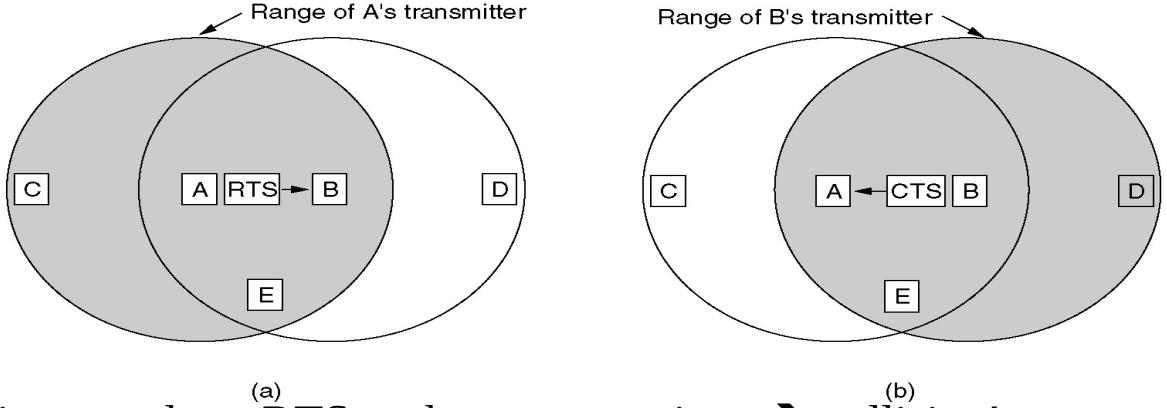



(b)

  - What matters is the status of receiver, not the transmitter.
    - Wireless LAN can only find out the status at transmitter

# PROTOCOLS: WIRELESS LAN

- **Multiple Access with Collision Avoidance (MACA)**
  - Basic idea: sender stimulates the receiver to output a short frame
    - Thus hosts around receiver will avoid transmission during the duration of the upcoming frame.
  - A sends RTS (Request To Send) to B: RTS contains length of data frame.
    - Hosts around A will not interfere with A
  - B replies CTS (Clear to Send): CTS also contains the data length
    - Hosts around B will remain silent during transmission.




  - Two stations send out RTS to the same receiver ➔ collision!
    - Back off a random time.

# OUTLINE

- Functions of MAC Sublayer
- MAC Protocols
- **Ethernet (IEEE 802.3)**
- Data Link Layer Switching
- Wireless LAN
- Broadband Wireless
- Bluetooth

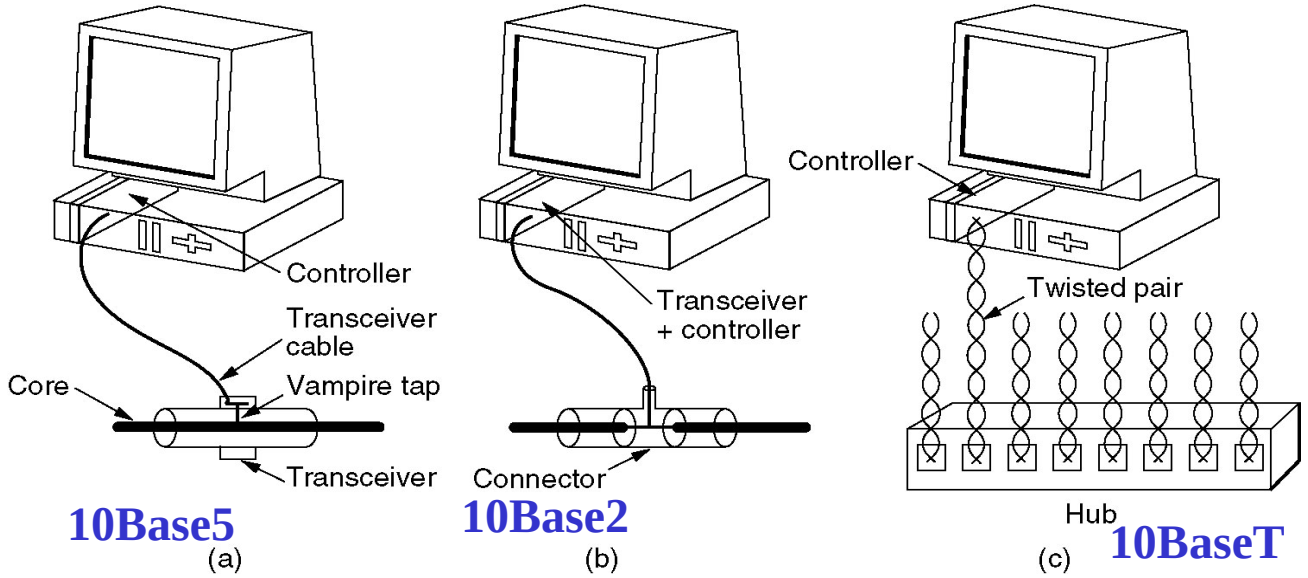
# ETHERNET: CABLING

- **Four types of cables**

| Name | Cable | Max. seg. | Nodes/seg. | Advantages |
|---|---|---|---|---|
| 10Base5 | Thick coax | 500 m | 100 | Original cable; now obsolete |
| 10Base2 | Thin coax | 185 m | 30 | No hub needed |
| 10Base-T | Twisted pair | 100 m | 1024 | Cheapest system |
| 10Base-F | Fiber optics | 2000 m | 1024 | Best between buildings |

  - First number: 10 ➔10Mbps
  - Base: baseband transmission.
  - Second number: 2 ➔ 200 meters, 5 ➔ 500 meters
  - Letter: T ➔ twisted pair, F ➔ Fiber Optic Cable

# ETHERNET: CABLING

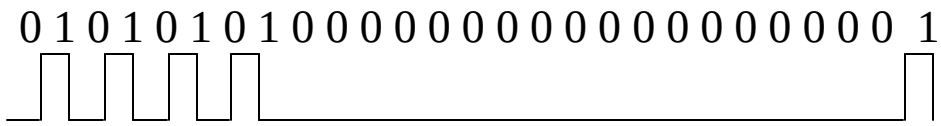



- **Transceiver**
  - handles carrier sensing and collision detection.
- **Controller**
  - Assembling data into frame
  - Computing checksum, etc.
- **Hub: named after the hub of a wheel.**
  - All the incoming cables are electronic connected. ➔ Physical layer device
  - No buffering, no processing.

# ETHERNET: MANCHESTER ENCODING

- **How to represent ‘0’ and ‘1’ in Ethernet**
  - Option 1: 5V represents ‘1’, 0V represents ‘0’
    - Cannot distinguish between ‘no data’ and ‘0’ ➔ no good!
  - Option 2: 5V represents ‘1’, -5V represents ‘0’
    - Different clock speed at transmitter and receiver will cause confusion about bit boundary
      - Especially after long runs of consecutive ‘1’s or ‘0’s.
    - Example:

**Sender:**

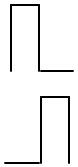



If the clocks at sender and receiver are not perfectly synchronized, receiver might not be able to count correctly how many ‘0’s are in the middle

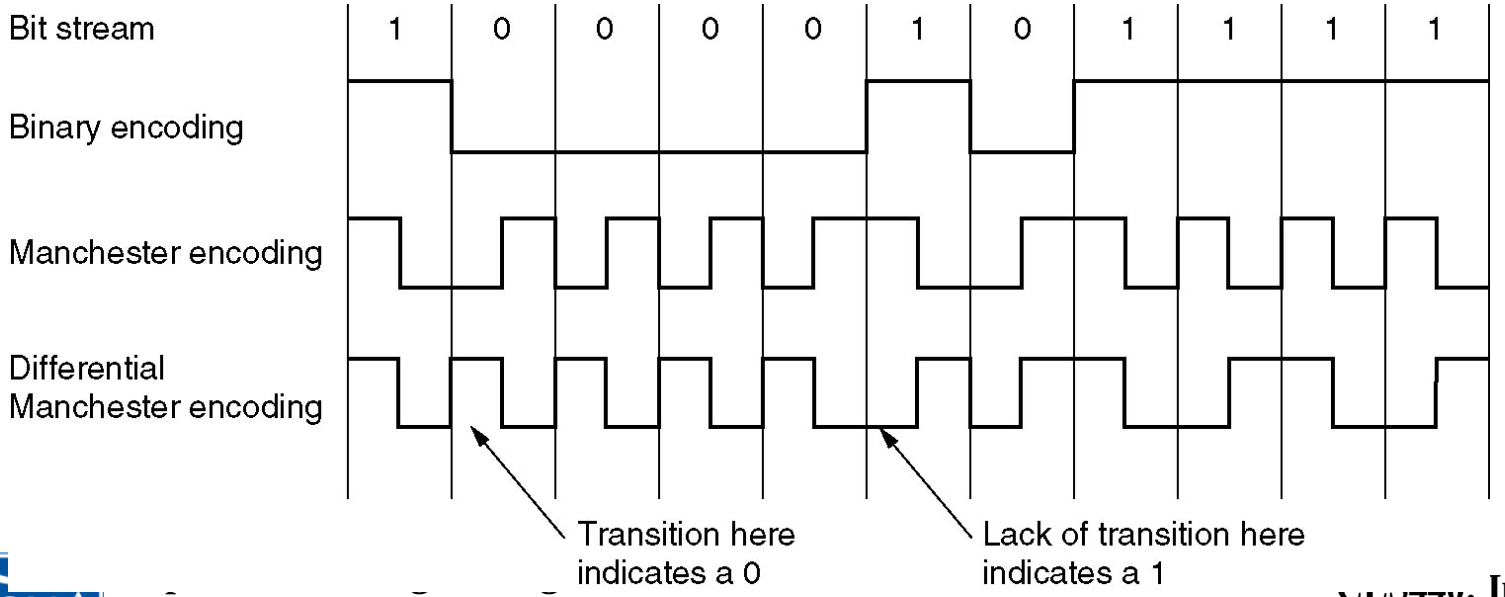
  - Can we carry clock information in the data?

# ETHERNET: MANCHESTER ENCODING

- **Solution: Manchester Encoding**
  - Each bit is represented by two Volts
  - "1": high voltage, then low voltage
  - "0": low voltage, then high voltage



- **Differential Manchester Encoding**
  - "0": presence of transition at the start of the interval.
  - "1": absence of transition at the start of the interval

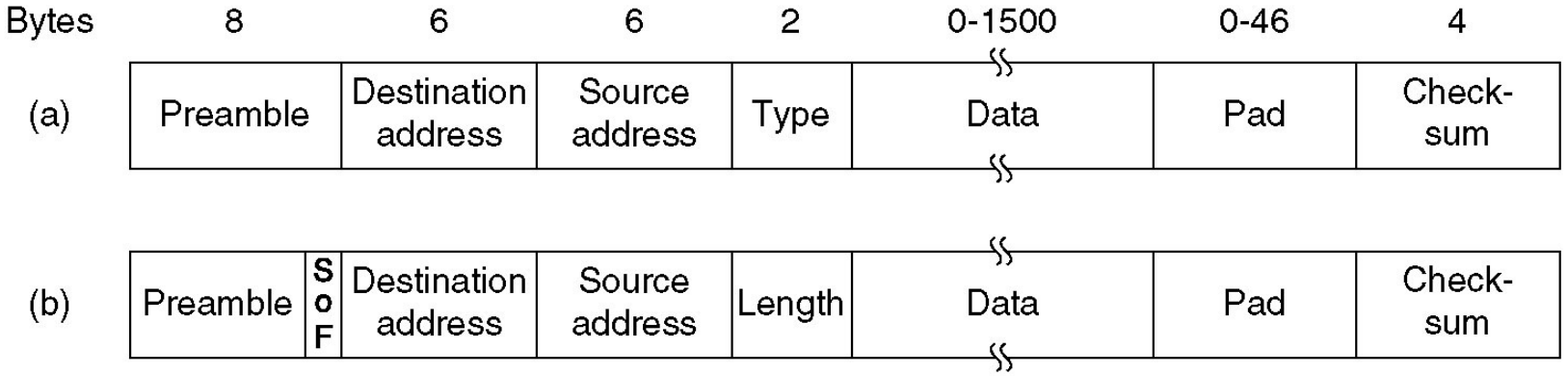


# ETHERNET: MANCHESTER ENCODING

- **Disadvantage:**
  - Signals change twice as fast as the original binary information ➔ signal bandwidth multiplied by two ➔requires two times the bandwidth.
- **Advantage:**
  - Always transitions in the middle of the bit
  - Receiver will not lose synchronization
- **IEEE 802.3**
  - Manchester encoding is used
  - High: 0.85V
  - Low: -0.85V
- **Differential Manchester encoding is used by IEEE 802.5 (token ring).**

# ETHERNET: MAC PROTOCOL

- **Frame structure**



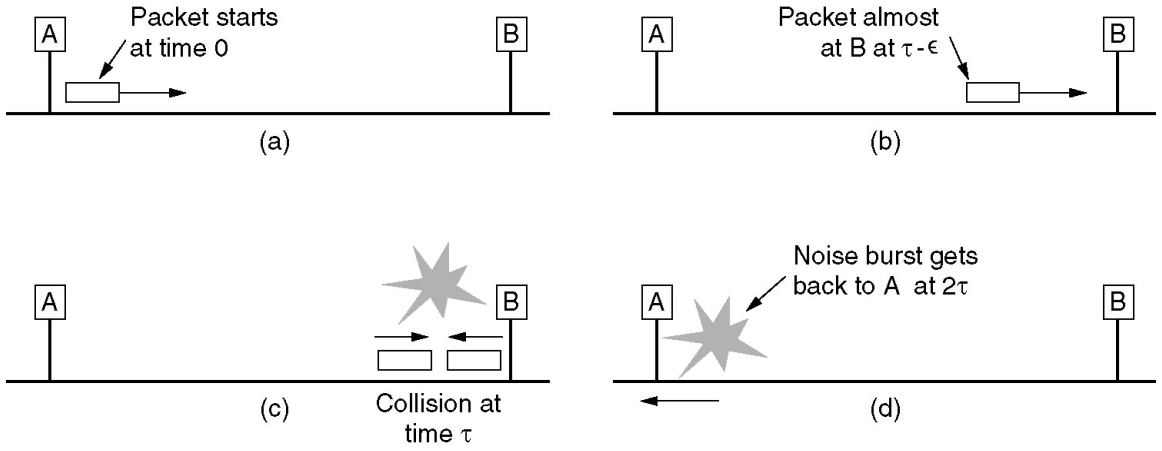

  - Preamble: 8 bytes, each byte containing the bit pattern 10101010
    - Allow receiver's clock to synchronize with sender's clock.
  - Address: 6 byte (MAC address)
    - E.g. 00-10-B5-0D-9E-B0 (Hexadecimal representations) 00000000-00010000-10110101-00001101-10011110-10110000
    - Bit 47: 0: ordinary address. 1: group address (multicasting, broadcasting)
      - All "1" address: broadcasting
    - Bit 46: 0: global address; 1: local address.
    - Totally $2^{48-2} \approx 7 \times 10^{13}$ global addresses!
      - Each network adapter has a unique global address, and is assigned by IEEE.

# ETHERNET: MAC PROTOCOL

- **Frame Structure (Cont'd)**
  - Type: tells the receiver what to do with the frame
    - E.g. tells kernel the frame belongs to which network layer protocol.
  - Length: the length of the frame.
    - The type information is carried by a small header of data field.
  - SOF (Start of Frame) byte: frame delimiter.
    - For compatibility with IEEE 802.4 and IEEE 802.5.
  - Checksum: CRC
  - Data: 0 ~ 1500 Bytes. (The limit is arbitrarily chosen)
  - Pad: make sure the frame is at least 64 bytes long.
    - Minimum length of Ethernet frame: 64 bytes. Why?

# ETHERNET: MAC PROTOCOL

- **Why minimum frame length?**




  - In the worst case scenario, it will take $2t$ for sender to detect collision.
    - $t$ : maximum propagation delay.
  - If a collision arrives after the frame has been completely sent out, the sender might think the frame has been transmitted successfully.
  - Make sure it will take more than $2t$ to transmit a frame!

# ETHERNET: MAC PROTOCOL

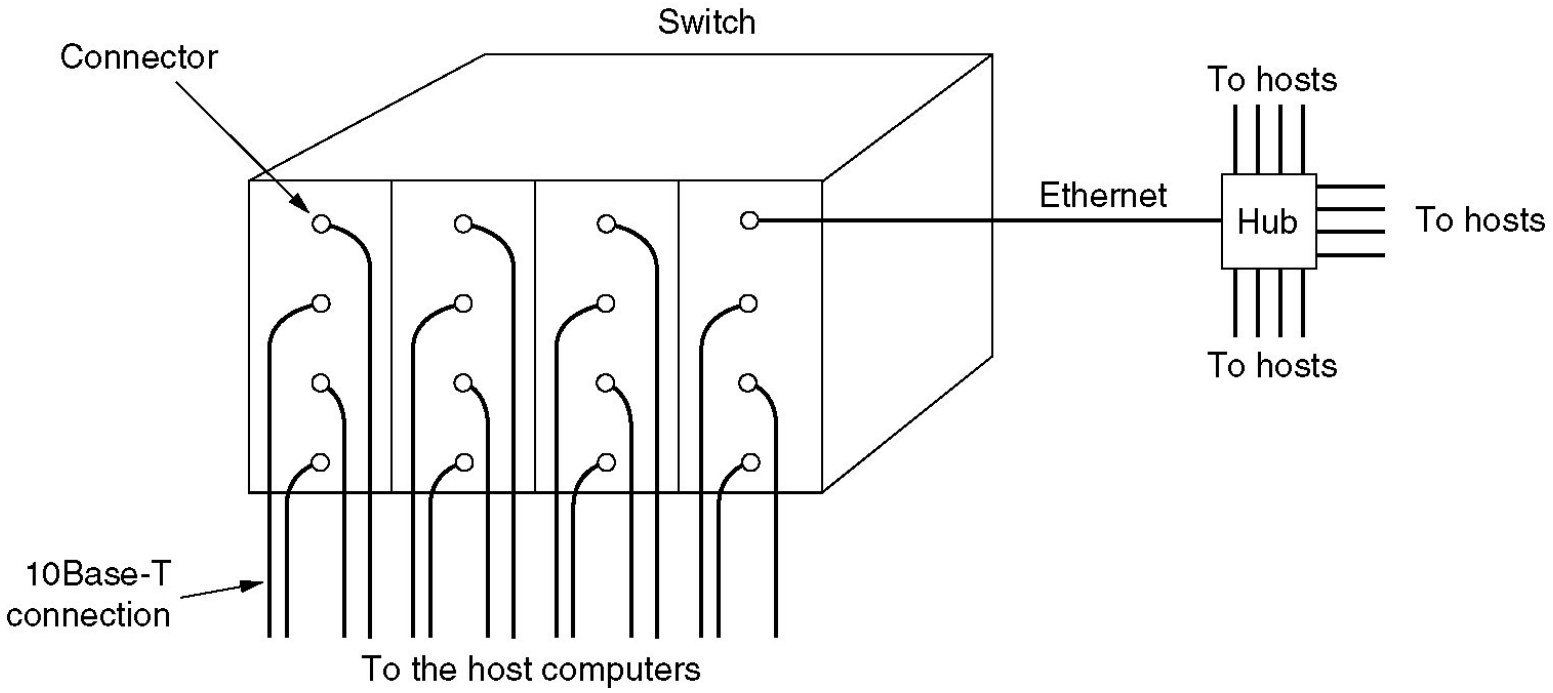
- **How to determine minimum frame length?**
  - 10Mbps LAN, length 2500 meters with four repeaters
    - $2t = 50ms$ for the worst case
    - 10Mbps ➔ Each bit takes 100 nsec = 0.1 micro-sec
    - $50ms / 0.1ms = 500$ bits
    - To add some margin of safety, the number is rounded to 512 bits ➔ 64 bytes!
  - 1Gbps (100 x 10Mbps), 2500 meter LAN
    - Minimum frame size = ?
  - 1Gbps, 250 meter LAN
    - Minimum frame size = ?

# ETHERNET: MAC PROTOCOL

- **Protocol used: CSMA/CD**
  - Contention period: short time slots, the length of each slot equals to the worst case round-trip delay $2t$ (or 512 bit time = $51.2ms$ ).
  - After a collision, wait for random time
    - how to determine the random time?
- **Binary Exponential Backoff**
  - After the first collision, each station waits either 0 or 1 slots before trying again.
  - After the second collision, each one picks either 0, 1, 2, or 3 before retrying.
  - After the *k*th collision, each one picks a random number between 0 and $2^k - 1$ slots.
  - After 10 collisions, the randomization interval is frozen at 1023.
  - After 16 collisions, error is reported.
  - Adaptively change the randomization interval!
    - Short random interval: shorter delay, larger chance of collision
    - Long random interval: longer delay, smaller chance of collision.

# ETHERNET: SWITCHED ETHERNET

- **Switch**
  - Contains 4 to 32 line cards, each line card has multiple ports.
  - Line cards are connected by high speed backplane.
    - Usually runs at Gbps, using proprietary protocol.



# ETHERNET: SWITCHED ETHERNET

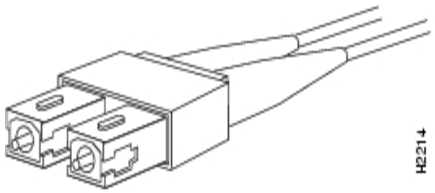
- **Two different types of switches**
  - Type 1: Each line card is an independent collision domain
    - Each line card is like a smaller LAN with CSMA/CD
      - Hosts connected to the same card could collision
      - One transmission per card
    - No collisions between cards
      - All cards can transmit parallel
  - Type 2: Each port is an independent collision domain
    - Each port is buffered, thus incoming frames can be buffered at on card RAM
      - All ports can transmit and receive at the same time!
    - No collision at all!

# ETHERNET: FAST ETHERNET

- **Fast Ethernet (IEEE 802.3u)**
  - Faster version of IEEE 802.3 ➔ 100 Mbps
  - Same frame formats, interfaces, procedural rules as IEEE 802.3
  - Just reduce the bit time from 100 nsec to 10 nsec
  - Backward compatible with IEEE802.3
    - Almost all switches can handle both 100Mbps and 10Mbps.

| Name | Cable | Max. segment | Advantages |
|---|---|---|---|
| 100Base-T4 | Twisted pair | 100 m | Uses category 3 UTP |
| 100Base-TX | Twisted pair | 100 m | Full duplex at 100 Mbps |
| 100Base-FX | Fiber optics | 2000 m | Full duplex at 100 Mbps; long runs |

- **100Base-FX**
  - Two multimode fibers
  - Each direction: 100Mbps



# ETHERNET: FAST ETHERNET

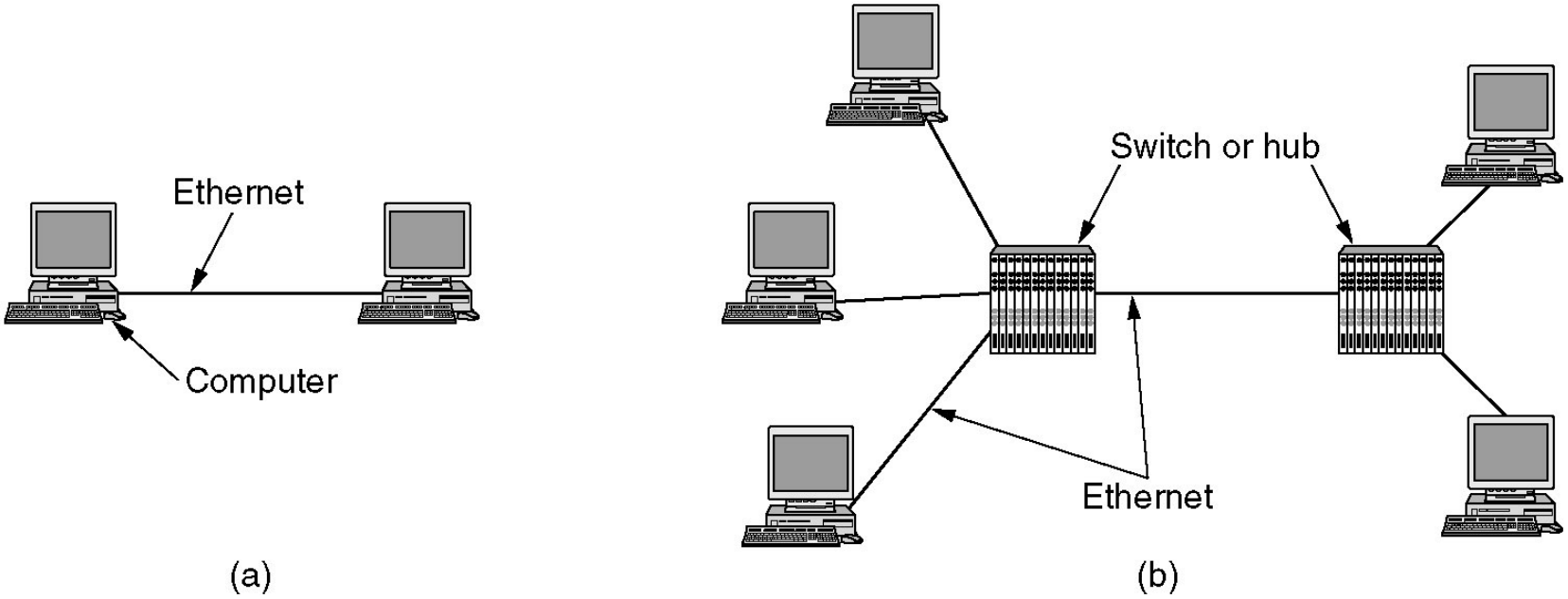
- **100Base-T4**
  - Four pairs of Category 3 UTP (Unshielded Twist Pair)
    - One pair: Host to Hub; One pair: Hub to Host.
    - Two other pairs: dynamically allocated to the two directions.
  - Clock speed: 25MHz (802.3: 20MHz)
  - Use ternary signals:
    - A single clock period, a wire can transmit either 0, 1, or 2.
  - Example: 3 pairs are used from Hub to Host. 1 pair from Host to Hub
    - At any single clock period, one of $3^3 = 27$ possible symbols can be transmitted.
    - 16 out of this 27 symbols are chosen to represent $\log_2 16 = 4$ bits of information.
    - 4 bit/clock; Clock speed: 25MHz. ➔
      - Data rate: 4x25=100 Mbps.
    - Host to Hub: 1 UTP, 100/3 = 33.3 Mbps
  - 4 bits are mapped to 3 trits ➔ 8 bits mapped to 6 trits ➔ 8B6T

# ETHERNET: FAST ETHERNET

- **100Base-TX**
  - Two pairs of category 5 UTP
    - One from Host to Hub; One from Hub to Host.
  - Clock speed: 125MHz
  - Coding scheme: 4B5B
    - Each clock period contains one binary signal
    - 5 clock periods: $2^5 = 32$ possible combinations (symbols)
    - 16 of the 32 combinations are chosen to represent $\log_2 16 = 4$ bits of info.
      - Some of the remaining 16 are used for control purpose and redundancy.
    - Mapping 4-bit of information onto 5-bit ➔ 4B5B
  - Data rate:
    - Each 5 clock periods contain 4-bit of information.
    - Duration of 5 clock periods: $t$ = 5 x 1/125MHz (sec)
    - Data rate: R = 4/$t$ = 4 x 125/5 = 100 Mbps

# ETHERNET: GIGABIT ETHERNET

- **Gigabit Ethernet (IEEE 802.3z)**
  - Same addressing (48-bit), frame format, min/max frame length as IEEE 802.3
  - Supports 1Gbps
  - Only supports point-to-point configuration ➔ no shared cable!



# ETHERNET: GIGABIT ETHERNET

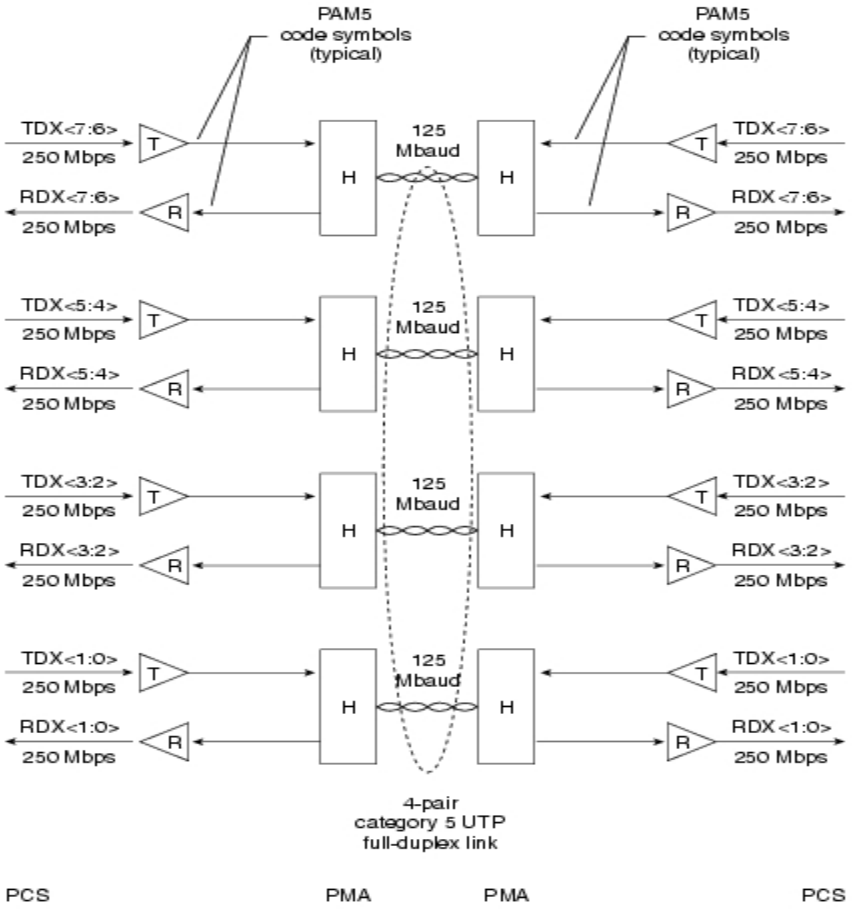
- **Two modes of operation**
  - Full Duplex
    - Can only be used by switches ➔ No collision ➔ doesn't need to worry about frame length and round trip delay.
  - Half Duplex
    - With Hubs ➔ Collision ➔ CSMA/CD
    - 64-byte frame ➔ 25 meters radius (802.3: 2500 meters)
    - Carrier Extension: hardware add padding frame to 512 bytes
      - Transparent to software
    - Frame Bursting ➔ transmit multiple frame in a single transmission

| Name | Cable | Max. segment | Advantages |
|---|---|---|---|
| 1000Base-SX | Fiber optics | 550 m | Multimode fiber (50, 62.5 microns) |
| 1000Base-LX | Fiber optics | 5000 m | Single (10 μ) or multimode (50, 62.5 μ) |
| 1000Base-CX | 2 Pairs of STP | 25 m | Shielded twisted pair |
| 1000Base-T | 4 Pairs of UTP | 100 m | Standard category 5 UTP |

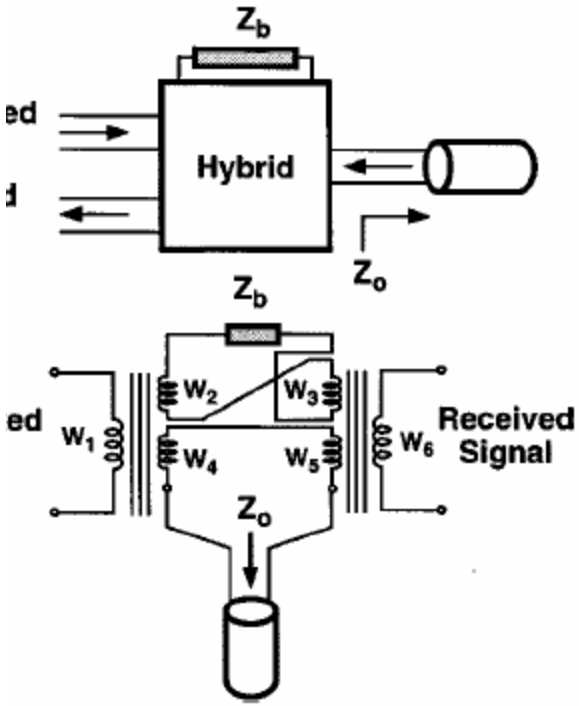
# ETHERNET: GIGABIT ETHERNET

- **Encoding:**
  - Fiber: 8B/10B
    - Map 8-bit of information on a 10-bit sequence
    - 10-bit sequence has $2^{10} = 1024$ possible combinations, how to choose 256 codeword:.
      - No codeword may have more than four consecutive identical bits
      - No codeword may have more than six 0s or six 1s.
  - 1000Base-T
    - 4 Category 5 UTP transmit simultaneously
    - Each clock period: send one of 5 voltage levels on each UTP
      - 2 bits of information (00, 01, 10, 11), and a special value for control
    - Each clock period: 8 bits of information of 4 UTPs
    - Clock speed: 125MHz ➔ data rate: 8 x 125 = 1000 Mbps = 1 Gbps
- **Flow control is supported by Gigabit Ethernet**
  - A special PAUSE frame is used to tell the sender pause for a certain time
    - Ethernet frame with type: 0x8808.

# ETHERNET: GIGABIT ETHERNET




H = Hybrid canceller transceiver
T = Transmit encoder
R = Receive decoder
Four PAM5 code symbols = One 4D-PAM5 code group



# ETHERNET: LOGICAL LINK CONTROL (LLC)

- **Logical link control (LLC): IEEE 802.2**
  - On top of MAC sublayer, provide service to network layer.
  - Support unreliable datagram, acknowledged datagram, and reliable connection oriented services
    - Error control, flow control (not supported by MAC sublayer)
  - Very similar to HDLC: seq#, ack#, etc.

